Hydrogen Sulfide Gas Sensor

SMD1007 Product Datasheet

1. Product Introduction

The SMD1007 hydrogen sulfide gas sensor is a MEMS micro gas sensor developed based on specific semiconductor gas-sensitive materials. It can be used to detect the hydrogen sulfide gas content in different scenarios.

The hydrogen sulfide gas sensitive material developed and made by IDM independently. When H2S gas contacts the sensitive material, the conductivity of the sensitive material will change. A specific test circuit can be used to convert the this change into an output voltage signal corresponding to the gas concentration.

2. Sensor Characteristics

- Small size
- Low power consumption
- High sensitivity
- Fast response and recovery time
- Simple test circuit
- Good stability
- Long life time

3. Main Application

It is widely used in the testing of hydrogen sulfide gas in scenarios such as

- Food testing
- Public sanitation
- Livestock
- Portable detectors

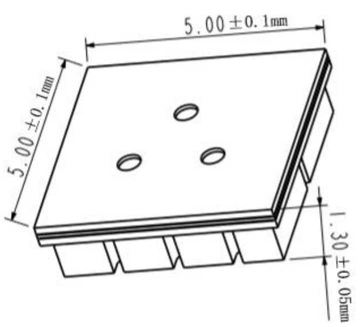
4. Product Description

4.1 Technical parameters

Table 1

Product Number			SMD1007
Product Type			MEMS Semiconductor Sensors
Standard Package			Ceramic Package
Target Gas			Hydrogen Sulfide H2S
Measurement Range			0 ~ 3 ppm (hydrogen sulfide)
Resolution			0.01 ppm
Standard Circuit Conditions	Sensor Voltage	V_C	5 V or 3.3 V DC
	Heating Voltage	V_H	1.8 V ± 0.05 V DC
	Load Resistance	R	Adjustable (subject to shipping sheet)
Gas Sensor Characteristics Under Standard Test Conditions	Heater Resistance	R_H	43 ± 5 Ω (measured at room temperature)
	Heating Power Consumption	P_H	≤ 36 mW
	Sensor Resistance	R_S	10 ~ 100 KΩ (tested in air)
	Sensitivity	S	R 0 (in air) / Rs (in 0. 2 ppm H2S)≥1.8
	Response Ratio	α	≤ 0.5 (R 1 pp m / R 0.2 pp m Hydrogen sulfide)
Standard Test Conditions	Temperature Humidity		20±2 ℃ 55±5 %RH
	Preheat Time		3-5 min
Response Time (T_{90})			< 3 0 s
Recovery Time (T_{10})			< 6 0 s
Life Time			≥ 3 years

4.2 Pin Definition Diagram



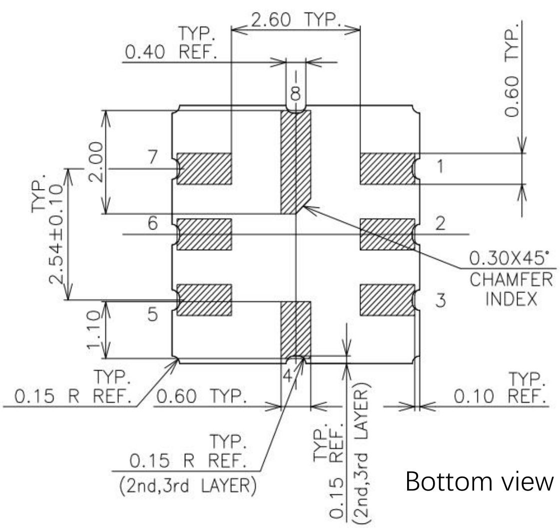
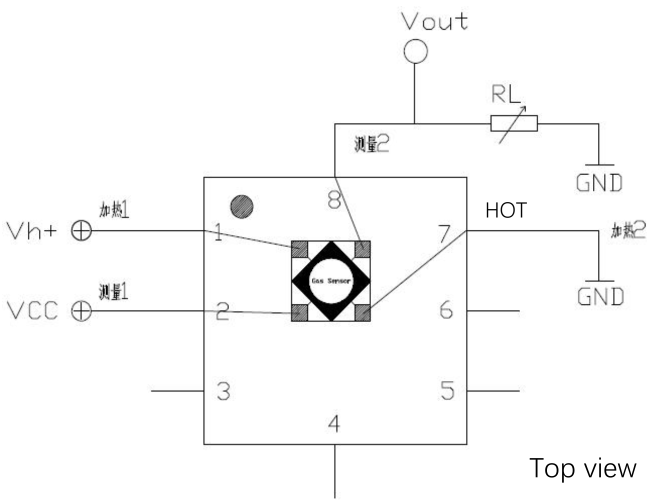


Table 2

Pin Number	Name	Definition
1	Vh+	Heating voltage supply
2	VCC	Sensor voltage supply
3	NG	/
4	NG	/
5	NG	/
6	NG	/
7	HOT	Heating GND
8	Vout	Sensor output voltage

4.3 Basic Circuit



Top view

Figure shows the basic test circuit for SMD1007 sensor. The sensor needs to apply 2 voltages: heater voltage and sensor test voltage. Vh+ is a specific voltage for the sensor heating element. Vcc is the sensor voltage. Vout is output test voltage which need connect the load resistor RL in series. These two voltage need set in right value according to datasheet, otherwise measurement performance will reduce and even damage the sensor.

5. Sensor Characterization

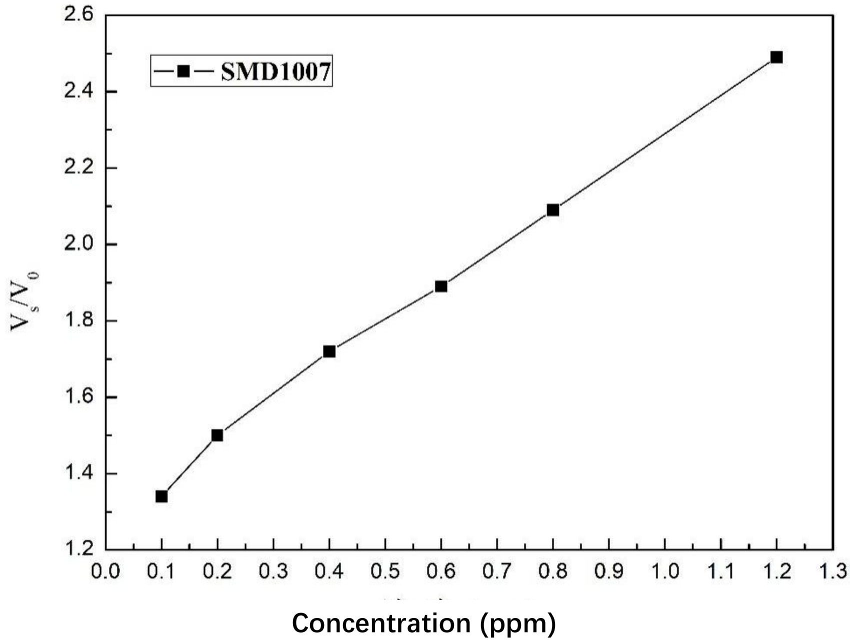


Figure 1 Sensor sensitivity characteristic curve

The test was completed under standard test conditions, and the load resistance used was 2 kΩ. In the figure, V0 represents the output voltage value to clean air, and Vs represents the output voltage value to hydrogen sulfide gas in different concentrations.

Note: in this section, if the unit is resistance, it is calculated by this formula:

$$R = \frac{R_L(V_{CC} - V_{OUT})}{V_{OUT}}$$

Each item definition refer table1

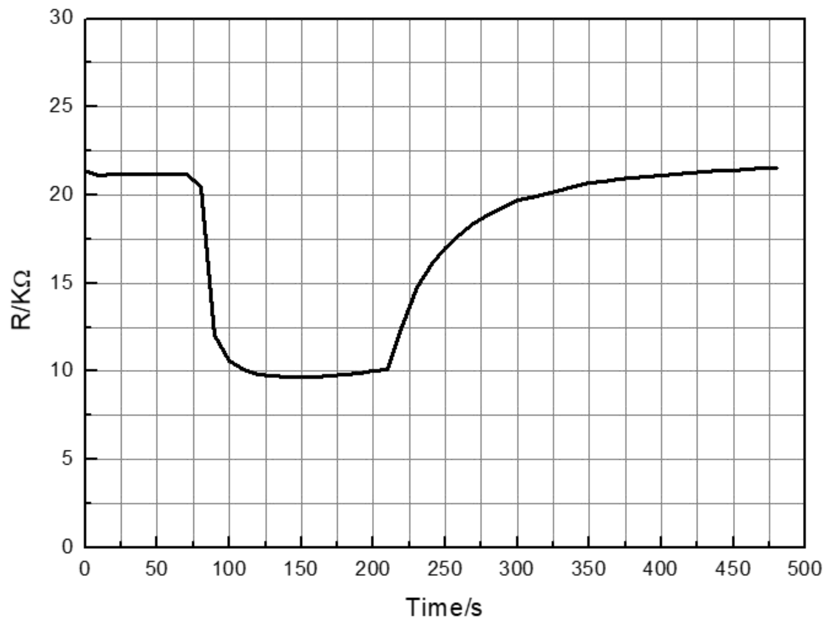


Figure 2 Sensor response-recovery characteristic curve

The curve in the figure shows the real-time voltage value read out from the sensor. The target gas is 0.2 ppm hydrogen sulfide. The test was completed under standard test conditions.

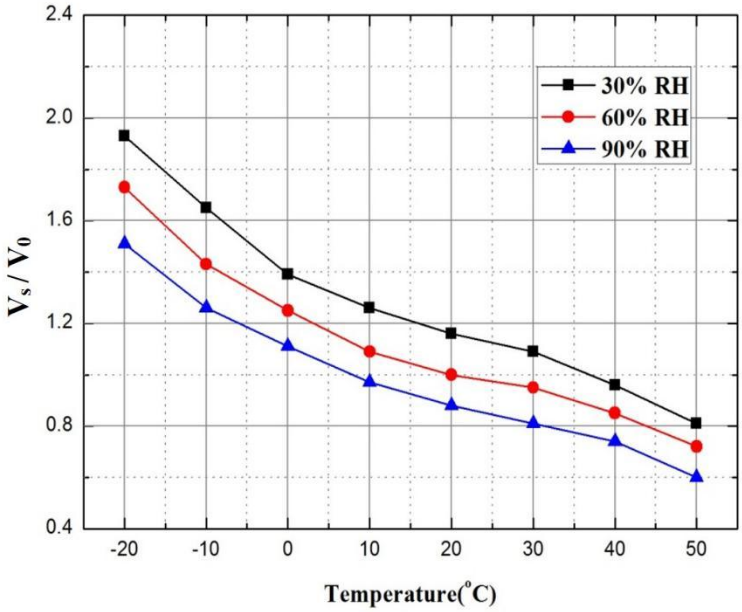


Figure 3 Sensor temperature and humidity characteristic curve

Above figure shows the temperature and humidity effect curves, Vs represents the response voltage in 0.2 ppm hydrogen sulfide gas under different temperature and humidity conditions, and V_0 represents the response voltage value in clean air under correspond temperature and humidity conditions.

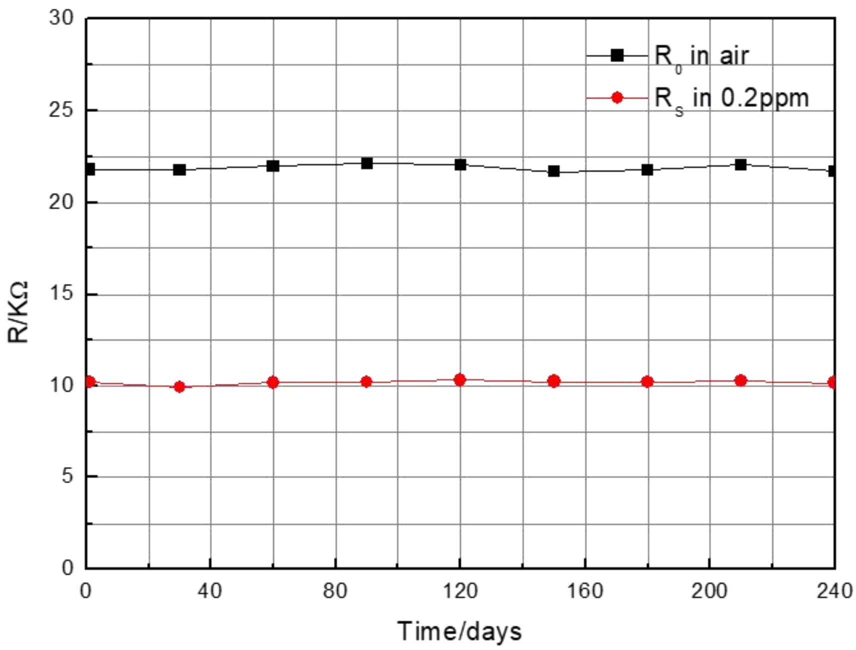


Figure 4 Long-term stability curve of the sensor

All tests in the figure were completed under standard test conditions. The horizontal axis is the continuous working time of the sensor, and the vertical axis is the resistance value of the hydrogen sulfide sensor. The test concentration is 0.2 ppm hydrogen sulfide. As represents the resistance value of the sensor in 0.2 ppm concentration, and R0 represents the resistance value of the sensor in clean air.

6. Product Shipping Packaging:

Generally shipped in carrier tape packaging. Customer can request other package when place the purchase.

Precautions:

1. Situations must be avoid

1.1 Exposure to volatile silicon compound vapor

The sensor should avoid being exposed to silicone adhesives, hair spray, silicone rubber, putty or other places where volatile silicone compounds exist. If the surface of the sensor is adsorbed with silicone compound vapor, the sensitive material of the sensor will be wrapped by silicon dioxide which formed by the decomposition of the silicone compound, which will degrade the sensitivity of the sensor and cannot be restored.

1.2 Highly corrosive environment

The sensor is exposed to high concentration of corrosive gas (such as H2S, SOX , Cl2 , HCl, etc.), it will not only cause corrosion or damage to the heating material and sensor leads, but also cause irreversible deterioration of the performance of sensing materials.

1.3 Pollution by alkali, alkali metal salts and halogens

The performance of the sensor may also deteriorate when it is contaminated by alkali metals, especially salt water spray, or exposed to halogens such as Freon.

1.4 Directly touch with water

Splashing or immersing the sensor in water will cause the sensor's sensitivity to decrease.

1.5 Freeze

If sensor surface appear with ice, it will cause sensing material crack and lost sensitive property.

1.6 Applied voltage is too high

If the voltage applied to the sensor or heater is higher than the specified value, even if the sensor is not physically damaged or destroyed, it may cause damage to the leads and/or heater and cause the sensor's sensitivity characteristics to degrade.

1.7 Voltage pins mismatch

If the wrong voltage is applied to the sensor or heater and signal pins, it may cause damage to the leads and/or heater and cause the sensor's sensitivity to deteriorate.

2. Situations need attention

2.1 Condensation

Under indoor use conditions, slight condensation will have small impact on the sensor performance. However, if water condenses on the surface of the sensitive layer and remains for a long time, the sensor characteristics may deteriorate.

2.2 In high concentration gas

Regardless of whether the sensor is powered on or not, long-term place in high-concentration gas will affect the sensor characteristics. For example, if the sensor directly be spray by lighter gas (butane), that will cause great damage to the sensor.

2.3 Long-term storage

If the sensor is stored for a long time without power, its resistance will have a reversible drift, which is related to the storage environment. The sensor should be stored in a sealed bag that does not contain volatile silicon compounds. After long-term storage, the sensor needs to be powered for a longer time before use to stabilize it. The storage time and corresponding aging time are recommended as following:

Storage time	Recommended preheating time
Less than 1 month	Not less than 6 hours
1-6 months	Not less than 12 hours
More than 6 months	No less than 24 hours

2.4 Long-term exposure to extreme environments

Regardless of whether the sensor is powered or not, to exposure in extreme conditions, such as high humidity, high temperature or high pollution for long time, it will seriously affect sensor performance.

2.5 Vibration

Frequently, excessively vibration may cause the sensor's internal leads to break. This type of vibration can be generated during transportation and by using pneumatic screwdrivers or ultrasonic welders on the assembly line.

2.6 Impact

If the sensor is subjected to a strong impact or falls, internal lead wires may break.

2.7 Conditions of assembling:

2.7.1 Manual welding is the most ideal welding method for sensors. The recommended welding conditions are as following:

Flux: Rosin flux with minimal chlorine content

Constant temperature soldering iron

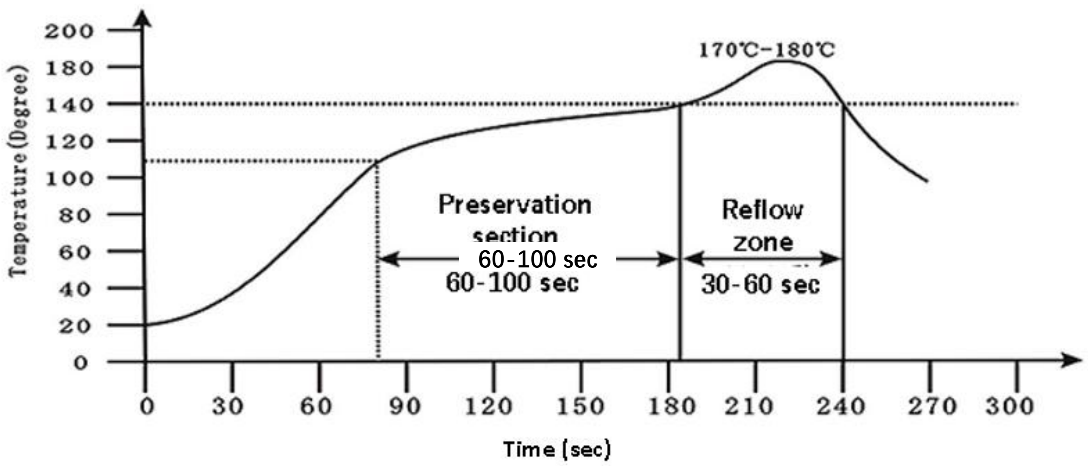
Temperature: 250℃

Time: no more than 3 seconds

2.7.2 The following conditions are recommended when using reflow soldering of SMT:

Solder paste: low temperature lead-free solder paste (Sn42Bi58)

The furnace curve is as following:



2.8 Anti-static

Anti-static bag packaging

Violation of the above usage conditions will degrade the sensor characteristics.

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